

Unsupervised Methods for High-dimensional data patterns

Marcos de Souza Oliveira

(presentation to ISG - CISUC)

Contextualization



resources consumption



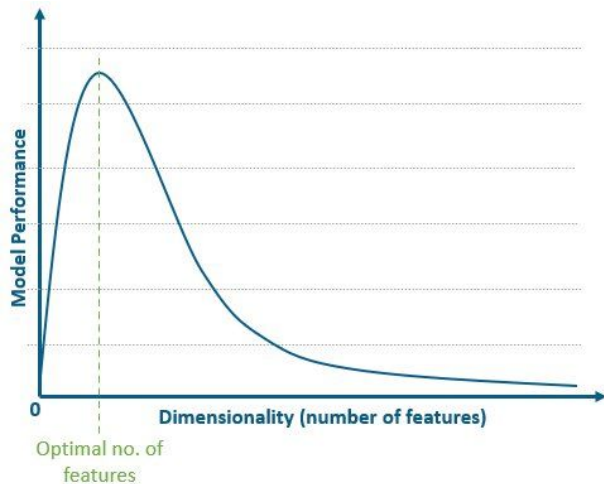
- Data labels can be scarce or expensive to curate
- Big Data require expensive infrastructure
- Limited number of samples (eg Bioinformatics)
- Deep-based methods require large train data
- Small data analysis
 - explainability
 - rationability
 - sustainability



*"Tradeoffs between quality and quantity require strategic **decision-making**, where small data often leads to faster, more accurate, and more cost-effective insights."*

Rather, I. H., Kumar, S., & Gandomi, A. H. (2024). Breaking the data barrier: a review of deep learning techniques for democratizing AI with small datasets. Artificial Intelligence Review.

Contextualization

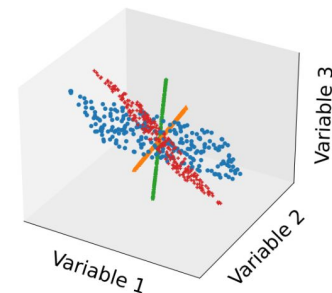


Hughes Phenomenon (+feat +obj)

ref:

<https://www.shiksha.com/online-courses/articles/curse-of-dimensionality/>

- Modern datasets exhibit considerable complexity, characterized by high dimensionality of data (Vidal, 2011)
 - Bioinformatics (eg genomics)
 - Computer vision (eg pixel-feature)
 - NLP (eg TF-IDF)
- Data patterns (eg clusters) may only be presented by subspaces



Approaches

- Unsupervised feature selection (UFS)
- Deep subspace clustering (SC)

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Unsupervised feature selection method based on iterative similarity graph factorization and clustering by modularity

Marcos de S. Oliveira ^{a b} , Sergio R. de M. Queiroz ^a , Francisco de A.T. de Carvalho ^a

^a Centro de Informática, Universidade Federal de Pernambuco, Recife, PE, Brazil

^b Recife Center for Advanced Studies and Systems (CESAR), Recife, PE, Brazil

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$$s_{a,b} = \begin{cases} 1, & \text{if } \mathbf{x}_a \in N_k(\mathbf{x}_b) \text{ or } \mathbf{x}_b \in N_k(\mathbf{x}_a) \\ 0, & \text{otherwise} \end{cases}$$

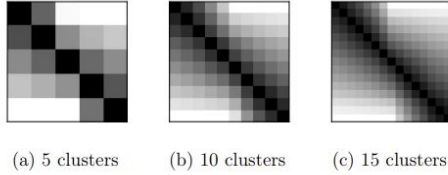


Fig. 16 – Three $\mathbf{D}$ distance matrices, representing data with 3, 10, and 15 balanced clusters (referring to the number of objects per cluster), respectively. The closer to black, the more similar and smaller the Euclidean distance.

Unsupervised feature selection method based on iterative similarity graph factorization and clustering by modularity, Expert Systems with Applications, 2022

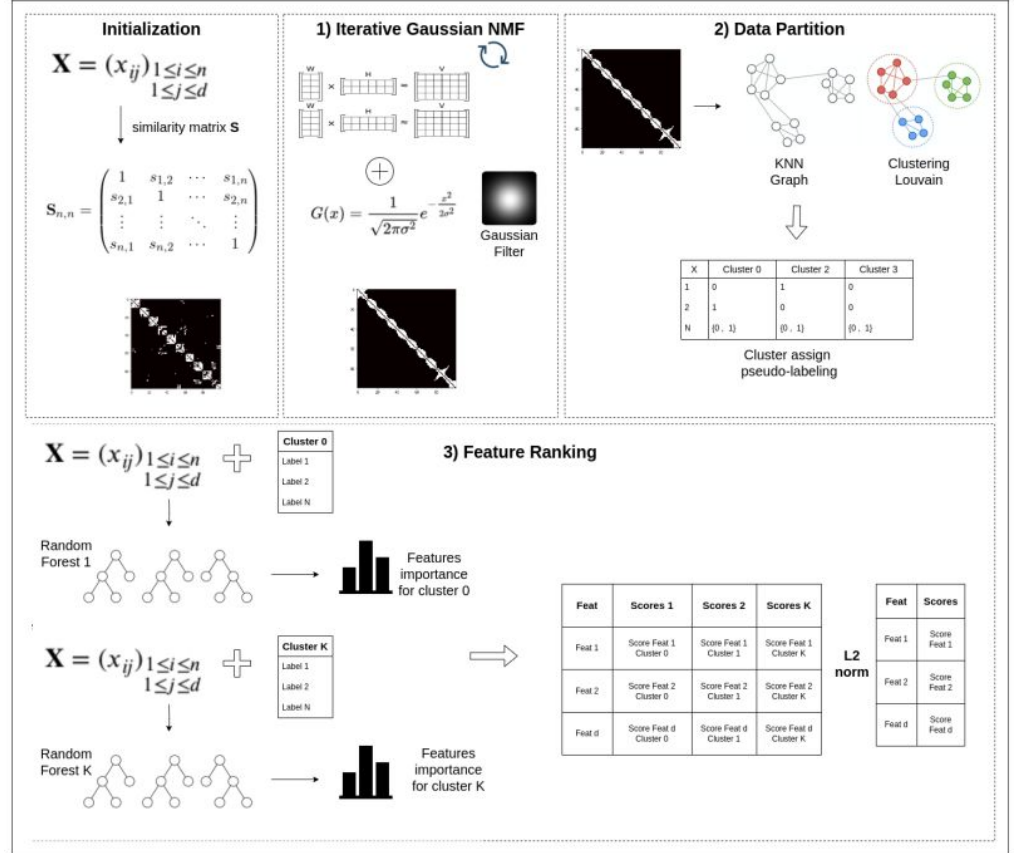


Fig. 7 – Workflow of the proposed UFS method.

Table 1 – Main properties of the state-of-art Methods

Method	Subspace Representation	Adaptive Local Structure Learning	Clustering Based	Feature Scoring Strategy
LS (He et al., 2006)	✗	✗	✗	Laplace
SPEC (Zhao and Liu, 2007)	✗	✗	✗	Spectrum
MCFS (Cai et al., 2010)	✗	✗	✓	Regression
UDFS (Yang et al., 2011)	✗	✗	✓	Eigenproblem
JELSR (Gu et al., 2011)	✗	✗	✗	Regression
NDFS (Wang and Zhang, 2012)	✗	✗	✓	Regression
RUFS (Qian and Zhai, 2013)	✓	✓	✓	Regression
GLSPFS (Liu et al., 2013)	✗	✓	✗	Regression
FSASL (Du and Shen, 2015)	✓	✓	✗	Regression
iDetect (Yao et al., 2015)	✗	✗	✗	Similarity
SCUFS (Zhu et al., 2017a)	✓	✓	✓	Regression
GSR_SF (Hu et al., 2017)	✓	✓	✗	Regression
LGRSFS (Zhu et al., 2017b)	✓	✓	✗	L2-norm
URAFS (Li et al., 2018)	✓	✓	✓	Regression
DGUFs (Guo and Zhu, 2018)	✗	✗	✓	L2-norm
AEFS (Han et al., 2018)	✓	✗	✗	Autoencoder
TSFS (Mirzaei et al., 2020)	✗	✗	✗	Deep Learning
SCFS (Parsa et al., 2020)	✓	✓	✓	Regression
Proposed	✓	✓	✓	Ensemble

Table 2 – Datasets

Dataset	Domain	Features	Instances	Classes
warpAR10P	Image	2400	130	10
warpPIE10P	Image	2420	210	10
pixraw10P	Image	10000	100	10
orlraws10P	Image	10304	100	10
nci9	Biological	9712	60	9
ProstateGE	Biological	5966	102	2
Carcinom	Biological	9182	174	11
TOX171	Biological	5748	171	4
ALLAML	Biological	7129	72	2
SMK-CAN	Biological	19993	187	2
isolet	Other	617	1560	26

Method	ACC Max	ACC Mean	NMI Max	NMI Mean	ARI Max	ARI Mean
LS	8,09	4,91	9,00	5,55	9,18	5,45
SPEC	8,18	5,09	9,00	6,18	9,09	5,64
MCFS	10,00	8,36	10,18	8,73	10,00	8,36
UDFS	8,55	8,91	7,45	8,36	8,82	10,09
JELSR	12,36	9,55	11,64	8,18	12,45	9,00
NDFS	8,73	10,55	12,73	12,64	13,09	13,09
RUFS	1,91	8,36	2,55	8,55	2,55	8,27
GLSPFS	8,18	8,09	9,27	7,45	8,27	7,27
FSASL	11,00	12,18	8,09	10,64	7,73	9,82
iDetect	7,36	12,27	8,55	11,91	9,18	12,27
SCUFS	5,55	8,55	6,18	9,8	4,91	8,91
DGUFs	12,09	9,91	11,91	10,00	11,91	10,18
AEFS	9,64	6,82	10,18	8,36	9,09	8,45
TSFS	12,09	9,36	10,36	7,73	10,18	7,18
SCFS	4,55	10,55	5,27	10,27	4,91	10,00
KNMFS	<u>3,64</u>	2,00	2,45	1,45	<u>3,27</u>	1,82



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Deep contrastive variational subspace clustering

Marcos de Souza Oliveira ^{a,b} , Sergio Ricardo de Melo Queiroz ^a , Cleber Zanchettin ^a ,
Francisco de Assis Tenório de Carvalho ^a

^a Centro de Informática, Universidade Federal de Pernambuco, Recife, Pernambuco, Brazil^b Intel Competence Center, Instituto Nacional de Telecomunicações - INATEL, Santa Rita do Sapucaí, Minas Gerais, Brazil

ARTICLE INFO

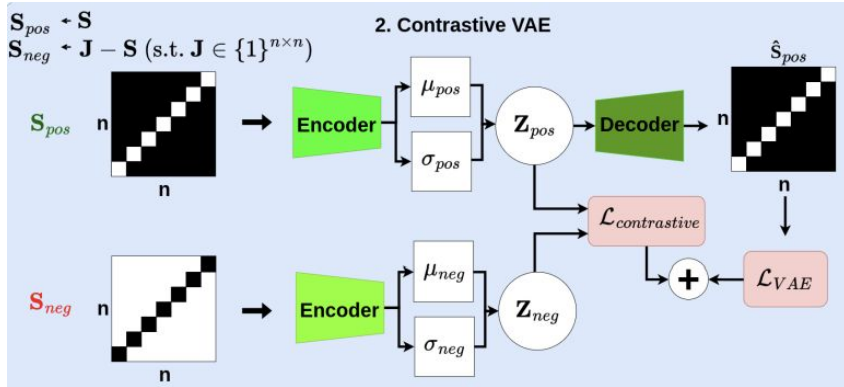
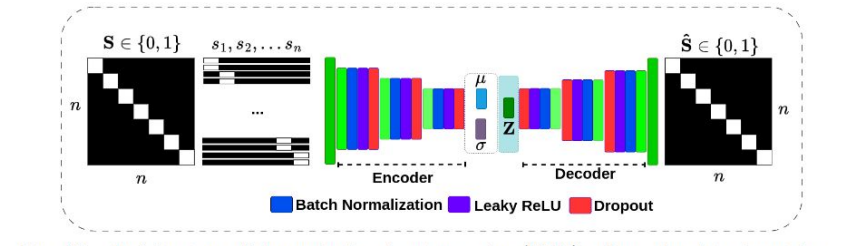
Communicated by J. Yu

Keywords:

Deep subspace clustering
Contrastive learning
Variational autoencoder
Neighborhood graph
Spectral clustering

ABSTRACT

Subspace clustering aims to group data points that lie in multiple low-dimensional subspaces within a high-dimensional space. Traditional shallow methods rely on linear optimization and self-expressive properties but struggle with complex, high-dimensional, and limited data samples. In contrast, Deep subspace clustering methods leverage neural networks to learn feature representations, but often specialize in specific domains, such as images, and require extensive training datasets. Furthermore, most approaches rely solely on raw features, making it difficult to capture meaningful structures in data with few examples. To address these challenges, we propose Deep Contrastive and Variational Subspace Spectral Clustering (DCSSC), a novel method that integrates relational information through a similarity matrix, enhancing clustering performance. DCSSC consists of three stages: (1) constructing a neighborhood graph for data representation, (2) training a contrastive variational autoencoder (cVAE) to learn embeddings, and (3) generating an enhanced distance matrix for spectral clustering. Our method is computationally efficient and performs well with minimal training data. Extensive experiments on simulated and 19 real-world datasets from diverse domains, including text, images, and bioinformatics, demonstrate the superiority of DCSSC over state-of-the-art deep and shallow clustering methods. Code is available at <https://github.com/marcosd3souza/DCSSC>.



$$\mathcal{L}_{cVAE}(S, \hat{S}) = \beta \cdot \mathcal{L}_{KL}(S, \hat{S}) + \mathcal{L}_e(S, \hat{S}) + \mathcal{L}_{contrastive}(S, 1 - S) \quad (4.3)$$

$$\mathcal{L}_{contrastive} = \sum_{i=1}^n [\max(0, \|z_{pos,i} - z_{neg,i}\|^2 - m) + \|z_{pos,i} + z_{neg,i}\|^2] \quad (4.2)$$

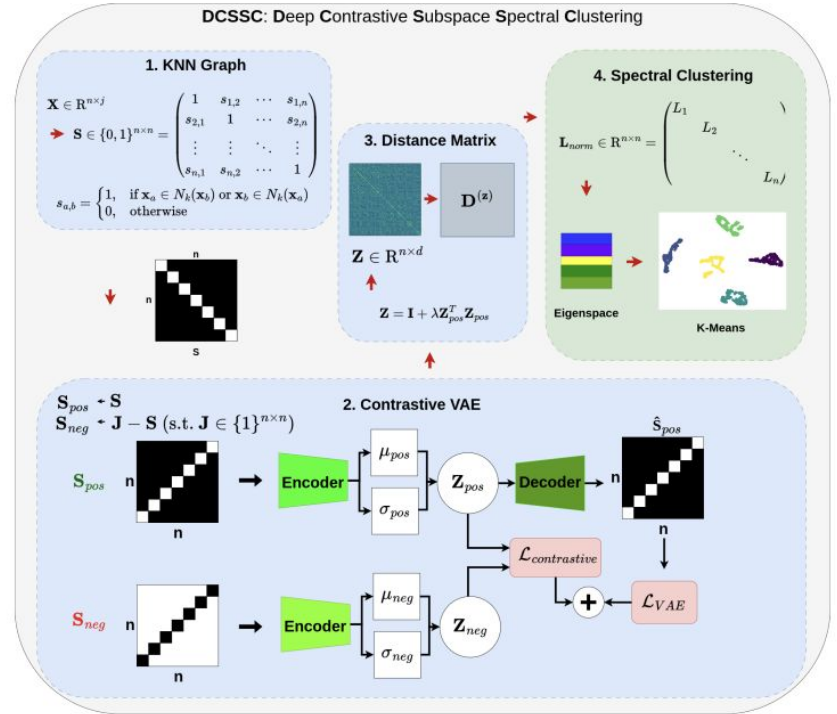


Fig. 18 – Workflow of the Deep Contrastive Subspace Spectral Clustering (DCSSC) method.

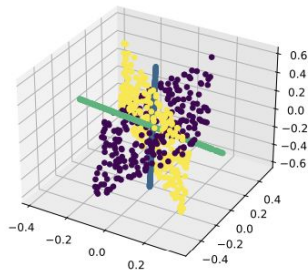


Fig. 19 – A toy example with 4 subspaces (clusters) by 1000 data examples and 3 dimensions (features).

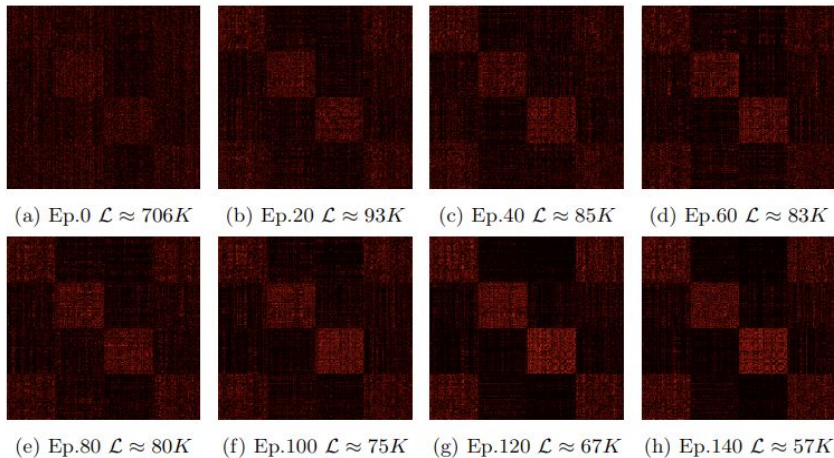


Fig. 20 – Visualization of the $\mathbf{S}$ matrix over the epochs for the synthetic dataset representing a subspace clustering (Figure 19). The $\mathbf{S}$ matrix is generated from the $\mathbf{Z}_{\text{pos}}$ latent space obtained from the cVAE, before regularization (4.4). Every 20 epochs, the matrix is updated, with $k = 50$ set to enhance the visibility of the block-diagonal structure.

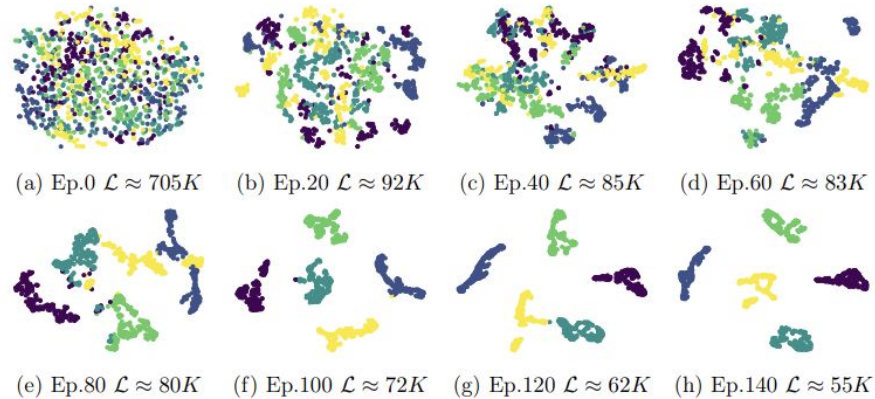


Fig. 21 – Visualization of T-SNE projection with two dimensions by $\mathbf{Z}_{\text{pos}}$ latent space obtained from the cVAE in a simulated dataset with 5 clusters (1000 examples, 10 features, dispersion of 0.5). The figure shows the separation of clusters over cVAE training epochs.

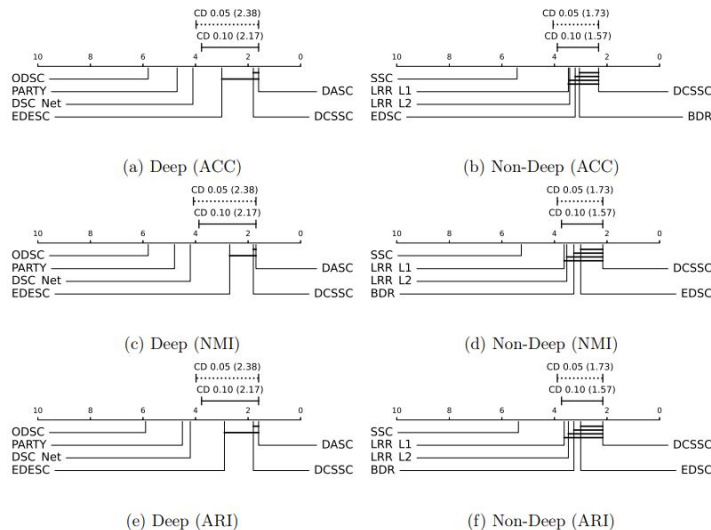


Fig. 24 – Critical Difference (CD) assessed using ACC, NMI, and ARI to compare the top-performing method with all other methods through the Nemenyi test.

Table 11 – Average rank of methods (lower is better) based on ACC, NMI, and ARI. Best results are highlighted in bold, and second-best are underlined.

Deep	DSC-Net	PARTY	DASC	ODSC	EDESC	DCSSC
ACC	4.10	4.70	1.60	5.80	3.00	<u>1.80</u>
NMI	4.20	4.80	1.70	5.80	2.70	<u>1.80</u>
ARI	4.20	4.50	1.60	5.90	2.90	<u>1.80</u>
Non Deep	LRR_L1	LRR_L2	SSC	EDSC	BDR	DCSSC
ACC	3.47	3.42	5.42	<u>3.21</u>	3.05	2.32
NMI	3.63	3.53	5.26	<u>3.00</u>	3.26	2.16
ARI	3.63	3.47	5.37	<u>3.00</u>	3.26	2.16

Table 6 – Datasets

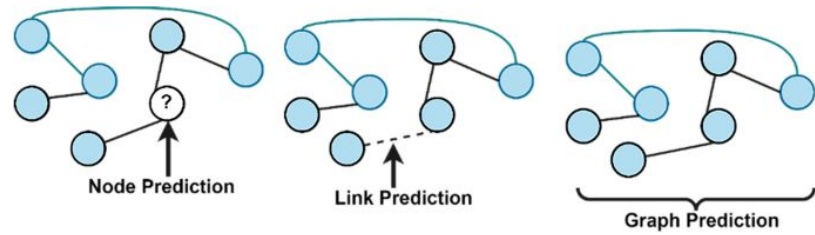
	Data	Obj.	Feat.	Domain	Size	Classes
1	USPS	9298	256	Image	16x16	10
2	CIFAR-10	1000	1024	Image	32x32	10
3	WarpAR10P	130	2400	Image	60x40	10
4	Pixraw10P	100	10000	Image	100x100	10
5	Yale	165	1024	Image	32x32	15
6	COIL20	1440	1024	Image	32x32	20
7	COIL100	7200	1024	Image	32x32	100
8	MNIST	4000	784	Image	28x28	10
9	ORL	400	1024	Image	32x32	40
10	YaleB	2414	1024	Image	32x32	38
11	ProstateGE	102	5966	Bio	–	2
12	TOX171	171	5748	Bio	–	4
13	lung	203	3312	Bio	–	5
14	lymphoma	96	4026	Bio	–	9
15	Carcinom	174	9182	Bio	–	11
16	RCV1	400	29992	Text	–	4
17	Reuters	900	18933	Text	–	9
18	TDT2	1900	36771	Text	–	19
19	Newsgroup20	2000	26214	Text	–	20

Dataset	LRR_L1	LRR_L2	SSC	EDSC	BDR	DSC_NET	PARTY	DASC	ODSC	EDESC	DCSSC
CIFAR10	40	40	1740	1	420	42	90	14	40	4	5
USPS	10	10	260	1	210	10	55	5	9	3	4
Yale	1	1	23	1	2	2	1	2	10	1	1
WarpAR10P	1	1	32	1	2	4	2	3	9	1	1
Pixraw10P	1	1	84	1	1	17	4	7	27	3	1
COIL20	32	90	7	1	840	75	227	27	53	5	9
MNIST	300	300	3360	5	13560	363	7767	56	177	13	65
ORL	4	4	150	1	9	8	6	5	11	2	1
YaleB	240	227	9840	2	3600	145	1300	67	107	9	24
COIL100	1230	1280	100180	33	64800	1231	74820	456	495	25	189
Carcinom	1	1	270	1	1	-	-	-	-	-	1
ProstateGE	1	1	44	1	1	-	-	-	-	-	1
TOX171	1	1	134	1	1	-	-	-	-	-	1
Lung	1	1	110	1	1	-	-	-	-	-	1
Lymphoma	1	1	17	1	1	-	-	-	-	-	1
newsgroup20	480	464	5410	6	2760	-	-	-	-	-	18
Reuters	30	30	432	1	240	-	-	-	-	-	5
TDT2	360	270	5440	7	2460	-	-	-	-	-	18
RCV1	6	6	102	1	40	-	-	-	-	-	1
Total	2740	2729	127635	67	88949	1897	84272	642	938	66	347

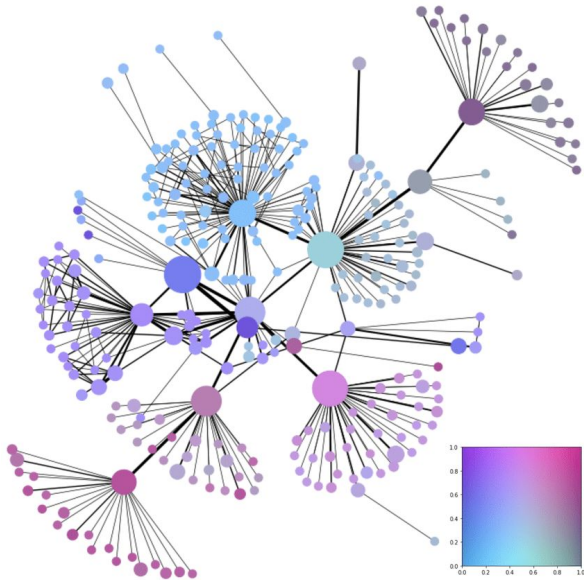
Table 7 – Total time spent (in seconds) by the methods considering a simple execution in all Datasets. The “-” indicates that the method is not applicable to the dataset’s domain.

Table 13 – Clustering results of the UFS method (KNMFS) and DSC method (DCSSC) both proposed on this research thesis measured by ACC, NMI and ARI. The results are presented in $\text{AVG} \pm \text{STD}$.

Datasets		Baseline	KNMFS	DCSSC	KNMFS + DCSSC
WarpAR10P	ACC	0.19 ± 0.01	0.31 ± 0.06	0.95 ± 0.03	0.92 ± 0.02
	NMI	0.19 ± 0.01	0.58 ± 0.06	0.93 ± 0.03	0.91 ± 0.02
	ARI	0.02 ± 0.01	0.37 ± 0.07	0.89 ± 0.06	0.85 ± 0.04
Pixraw10P	ACC	0.94 ± 0.01	0.95 ± 0.04	0.90 ± 0.02	0.91 ± 0.03
	NMI	0.94 ± 0.01	0.96 ± 0.02	0.87 ± 0.02	0.89 ± 0.03
	ARI	0.87 ± 0.01	0.92 ± 0.04	0.79 ± 0.03	0.81 ± 0.06
ProstateGE	ACC	0.03 ± 0.01	0.62 ± 0.01	0.98 ± 0.01	0.94 ± 0.05
	NMI	0.03 ± 0.01	0.05 ± 0.02	0.88 ± 0.01	0.90 ± 0.03
	ARI	0.02 ± 0.01	0.05 ± 0.01	0.92 ± 0.01	0.86 ± 0.09
TOX171	ACC	0.25 ± 0.01	0.42 ± 0.03	0.89 ± 0.01	0.93 ± 0.04
	NMI	0.25 ± 0.01	0.27 ± 0.08	0.77 ± 0.01	0.87 ± 0.06
	ARI	0.20 ± 0.01	0.14 ± 0.05	0.72 ± 0.01	0.83 ± 0.09
Carcinom	ACC	0.66 ± 0.01	0.72 ± 0.06	0.69 ± 0.01	0.84 ± 0.17
	NMI	0.58 ± 0.11	0.70 ± 0.06	0.81 ± 0.01	0.81 ± 0.13
	ARI	0.52 ± 0.01	0.54 ± 0.08	0.62 ± 0.01	0.73 ± 0.22



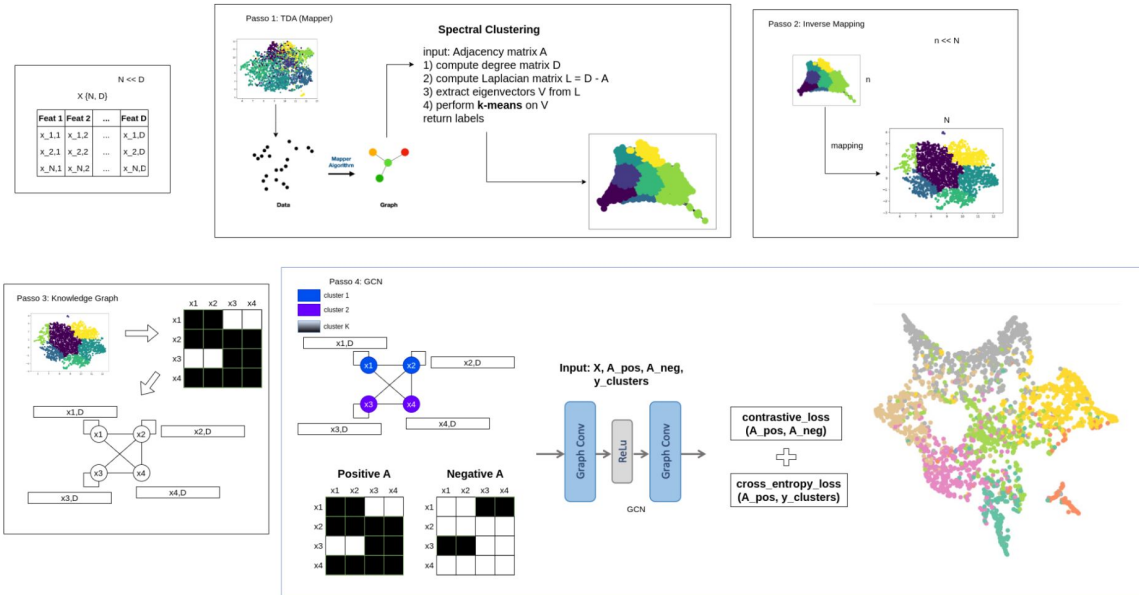
Future/Ongoing Works (GNN)



Graph	Number of nodes	Number of edges
Cora	3 327	4 732
Citeseer	2 708	5 429
Pubmed	19 717	44 338

Proposed Works (methods)

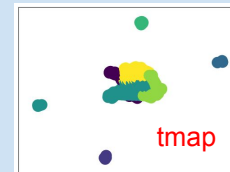
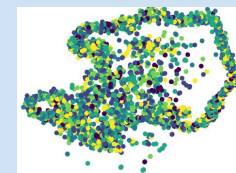
Topological Deep Graph Clustering



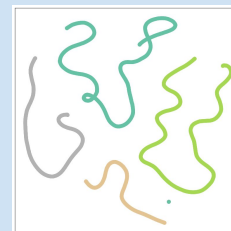
Work in progress

<https://github.com/marcosd3souza/TDGC>

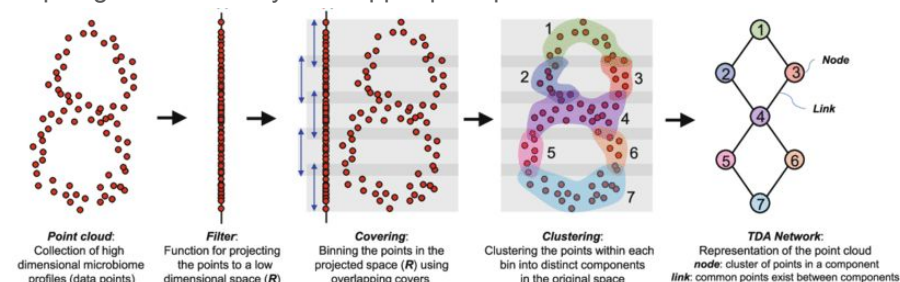
Citeseer (6 classes)
without contrastive term yet



GCN with **y tmap**
Epoch: 001, Loss: 1.9440
Epoch: 099, Loss: 0.0780



Topological Data Analysis: Mapper principle



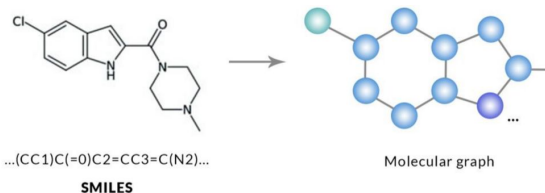
Liao, T., Wei, Y., Luo, M. et al. (2019). tmap: an integrative framework based on topological data analysis for population-scale microbiome stratification and association studies. Genome Biology.

Proposed Works (methods)

Deep Graph Clustering - SMILES

- ReLeaSE dataset

- 10K molecules
- **target:**
 - logP: measure of how a molecule interacts with different solvents
 - pIC50: Biological target or enzyme
- SMILES notation to graph



Work in progress
improvements: include supervision
[colab code here](#)

Integration of the natural data patterns, or clusters, alongside human expertise (supervision) can enhance data understanding and efficient AI modelling

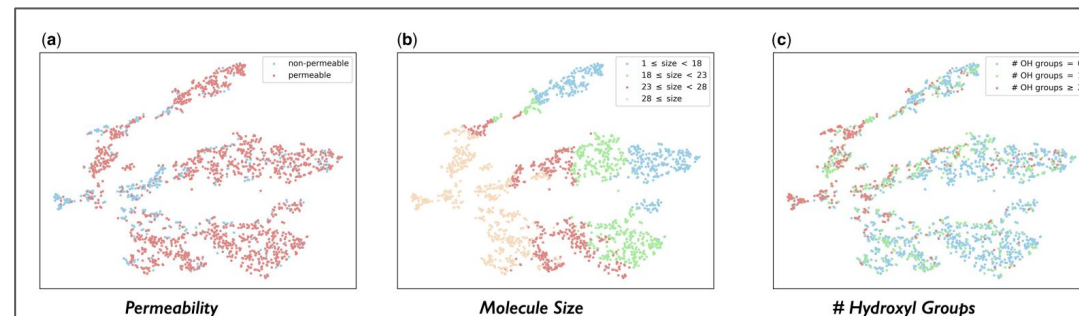
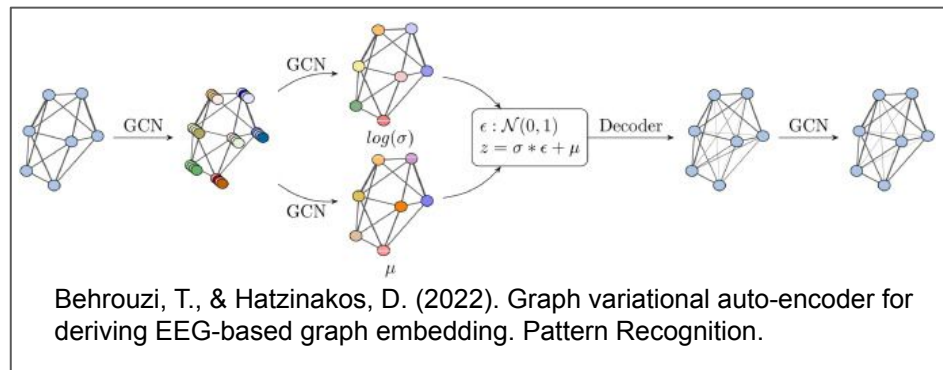
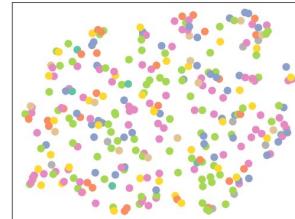
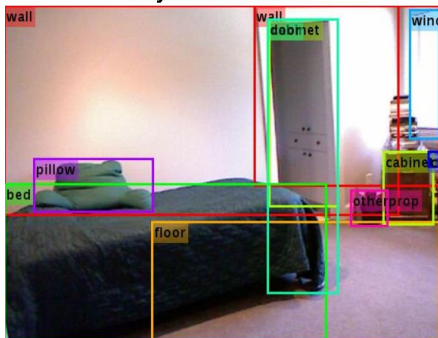


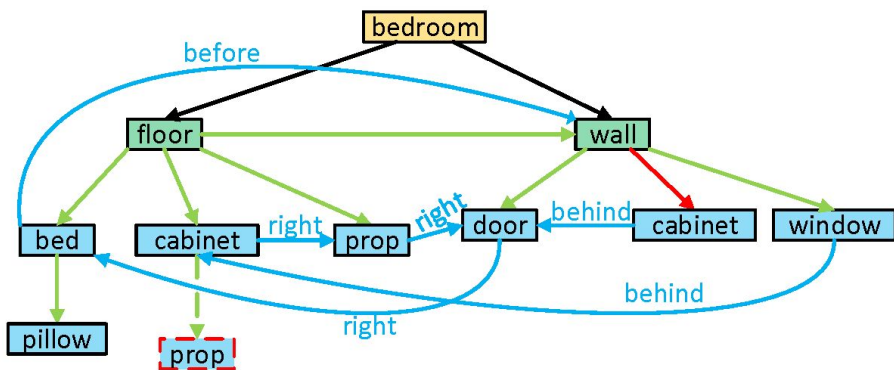
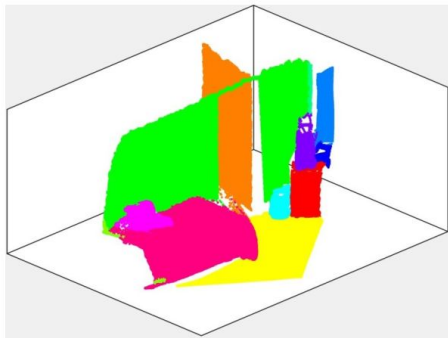
Figure 6. Visualized molecule embeddings on the BBBP dataset.

Li, A., Casiraghi, E., & Rousu, J. (2024). Chemical reaction enhanced graph learning for molecule representation. Bioinformatics.

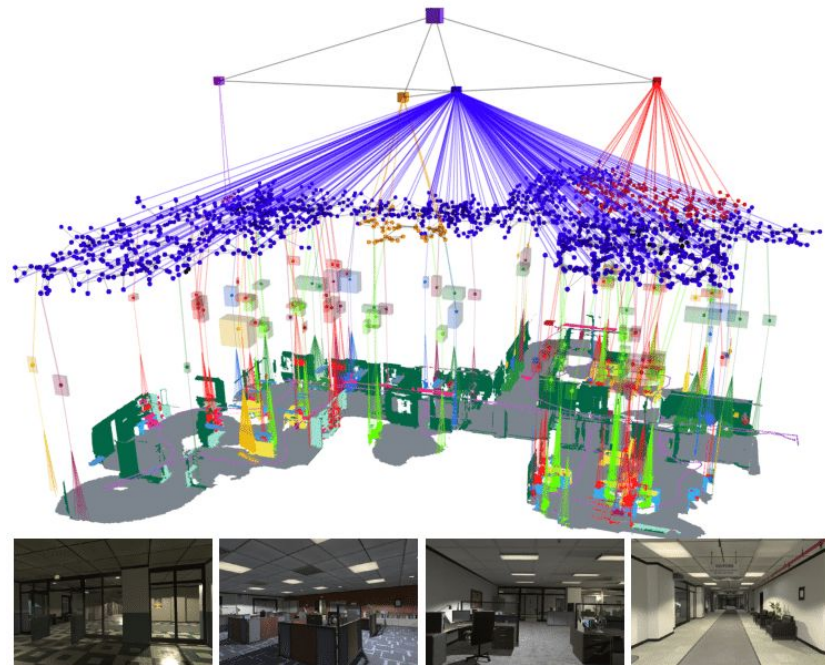
Object detection



3D cloud



Yang, M. Y., Liao, W., Ackermann, H., & Rosenhahn, B. (2017). On support relations and semantic scene graphs. *ISPRS journal of photogrammetry and remote sensing*, 131, 15-25.



Hughes, N., Chang, Y., & Carlone, L. (2022). Hydra: A real-time spatial perception system for 3D scene graph construction and optimization. *arXiv preprint arXiv:2201.13360*.